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Localization-Driven Multi-Robot Exploration for Indoor Search and Rescue

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ABSTRACT Search and Rescue (SAR) operations in complex indoor environments require efficient robotic coordination to ensure timely victim detection while optimizing area coverage and operation time. However, relying solely on exploration may be insufficient, as early victim localization can enhance path planning and prioritize victim access over the exploration of empty areas. In this work, we integrate Radio Frequency (RF)-based victim localization into State-of-Art multi-robot exploration approaches to guide decision-making and prioritize areas with higher likelihood of victim presence. Specifically, we perform probabilistic victim localization using Angle of Arrival (AoA) triangulation of RF signals, enhanced by Received Signal Strength Indicator (RSSI) filtering. This information is then used to dynamically update exploration priorities and replan robot paths toward potential victim locations. We evaluate the proposed method using the Cramér-Rao Lower Bound (CRLB) and Root Mean Square Error (RMSE). Extensive simulations confirm that the proposed approach significantly increases SAR efficiency, allowing up to $\sim 34\%$ of victims to be reached earlier in typical indoor environments and reducing the total mission time by up to 50%.

INDEX TERMS Robotics, Autonomous Mobile Robot (AMR), collaborative mobile robots, localization, exploration, Integrated Sensing and Communication (ISAC), Search and Rescue (SAR).

I. INTRODUCTION

TRADITIONAL Search and Rescue (SAR) operations that heavily rely on human intervention present inherent limitations, especially in hazardous or inaccessible environments. Autonomous robotic solutions have therefore emerged to improve SAR efficiency by enabling rapid and systematic exploration while minimizing risks to human first responders [1]. In this context, real-time understanding of the environment is essential, as well as estimating the robot's own position within it [2]. This process, known as Simultaneous Localization And Mapping (SLAM), is a foundational capability for autonomous navigation in unknown and unstructured indoor environments. SLAM enables robots to maintain spatial awareness, avoid obstacles, and plan efficient paths, all of which are critical for timely and effective rescue operations [3].

While SLAM is fundamental for localization and mapping, it does not inherently address how to guide exploration to locate victims [4]. Exploration strategies build upon SLAM by introducing decision-making logic that determines where the robot should move next. They open to incorporate forms of area prioritization and multi-step planning, where robots plan across sequences of goals rather than focusing on the next single-step action.

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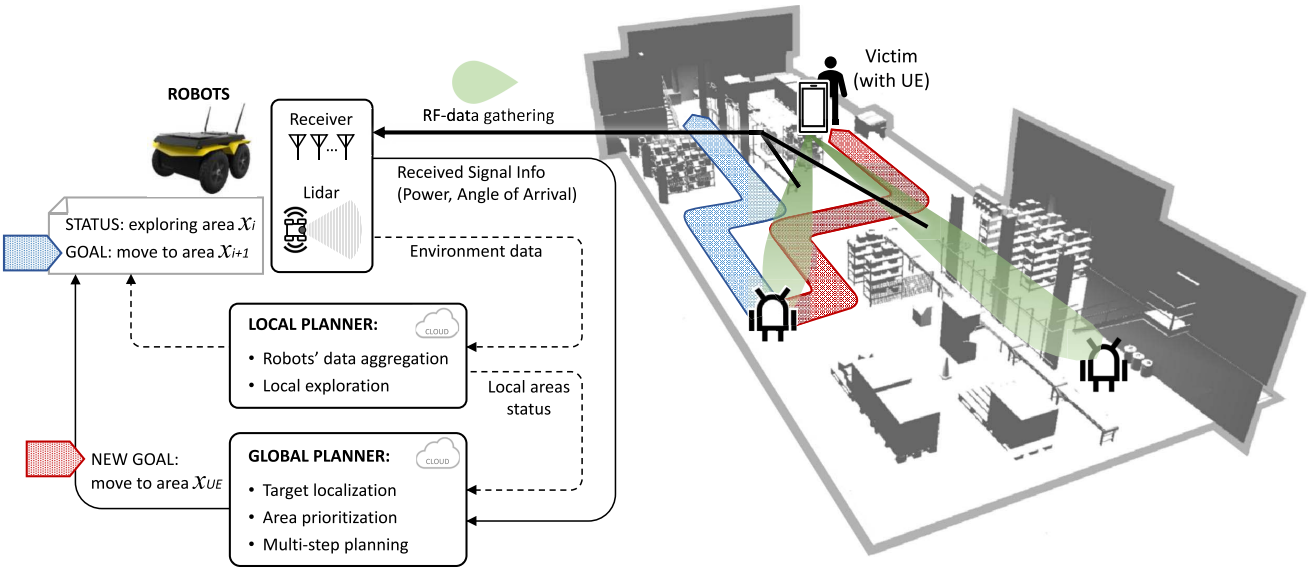


FIGURE 1. SAR operation in an indoor scenario. The figure illustrates the concept of path re-planning after victim localization. Robots collect RF information, which is processed by a planner to generate a new goal.

State-of-Art (SoA) exploration approaches include these features, implementing global progress tracking through abstractions such as graphs or grids that represent navigation status and constraints [5]. Although these features are essential to reach victims located away from current exploration frontiers, the original system lacks the ability to detect and respond to remote victim signals. In such scenarios, dynamically adjusting the exploration trajectory based on evolving mission information and priorities is crucial, particularly when remote localization signals (e.g., SOS signal transmissions) become available [6]. Such inputs provide valuable cues that can inform strategic planning and re-prioritization of goals during the mission.

Although multi-robot indoor exploration for SAR has seen substantial progress, existing methods still fail to incorporate remote victim localization into the autonomous decision-making process explicitly. In this work, we extend traditional goal-driven exploration approaches, typically considered in robotic-based SAR operations, by integrating remote victim localization as a new input for global planning. Specifically, we employ real-time remote sensing mechanisms based on established Received Signal Strength Indicator (RSSI) and Angle of Arrival (AoA) triangulation methods. The proposed framework is designed to be agnostic to specific frequency bands or propagation environments, thereby supporting a broad range of applications from indoor robotics to large-scale outdoor search and rescue. However, in this work, we select, instantiate, and evaluate the framework in the millimeter-wave (mmWave) regime, which offers high temporal resolution, strong spatial directivity, and reduced multipath scattering [7]. These enhancements enable the explorer to identify and prioritize victim locations during ongoing exploration, allowing the system to reallocate exploration effort toward areas of high rescue relevance dynamically (Fig. 1). To that end, we implement and tune a mathematical model for probabilistic victim localization and integrate this information as a new input to guide exploration toward high-priority targets.

The main contributions of this paper are:

1) We integrate real-time victim localization into a multi-robot exploration strategy to define prioritized goals and to enable consequent path replanning.
2) We develop a RF-based victim localization module that combines RSSI filtering with AoA triangulation.
3) We evaluate the accuracy of the probabilistic victim positioning using the Root Mean Square Error (RMSE), comparing it against the Cramér-Rao Lower Bound (CRLB), to relate empirical results to theoretical estimation limits.
4) We evaluate the enhanced system through extensive simulations, demonstrating up to $\sim 34\%$ improvement in early victim reach and up to 50% reduction in total operation time when victims are located in areas that would be explored late in SoA approaches.

The remainder of this paper is organized as follows: Section II reviews related works on RF localization and autonomous exploration. Section III highlights the importance of victim localization for improving SAR mission effectiveness. Section IV describes the proposed localization model. Section V presents the overall system architecture and its core modules. Section VI details the simulation scenarios and the ray-tracing-based RF environment model. Section VII introduces performance metrics and evaluates system performance through simulation. Finally, Section VIII concludes the paper with a discussion of limitations and future work.

II. RELATED WORKS

Robotics has become essential in SAR operations, particularly in hazardous and hard-to-reach scenarios, or post-disaster environments where a lack of prior knowledge about the area poses significant challenges [8]. Among robotic platforms, Unmanned Aerial Vehicle (UAV) deployment has long been a topic of interest in the research community [9], with foundational works dating back to 2010s [10], [11] and even earlier [12], [13]. More recently, there has been increasing interest in integrating RF-based techniques into SAR frameworks, particularly for victim localization in outdoor environments. The framework proposed in [14], dubbed as SARDO, utilizes an IMSI-catcher and Machine Learning (ML)-enhanced pseudo-trilateration to locate individuals in areas where cellular networks are not operational. 3DSAR+ [15] applies ML techniques to improve User Equipment (UE) localization accuracy, while [16] addresses the challenge of locating victims when devices may run out of power. Other studies focus on active signaling from the victim, where the user's device acts as the guiding factor [17]. In [6], UAVs-aided exploration for victim localization is guided by Reinforcement Learning (RL) agents fed with RSSI signal variations. Reference [18] proposes a post-earthquake localization approach that leverages RSSI signals from buried cell phones, where signal fluctuations are mitigated through hybrid filtering and processing to improve positioning accuracy. In other scenarios, a victim's smartphone may automatically activate Bluetooth Low Energy (BLE) following a disaster, allowing rescuers to perform RSSI measurements for victim trilateration [19].

These works demonstrate the potential of combining wireless signals and data-driven models to infer victim positions in open outdoor settings. However, indoor scenarios involving mobile ground robots have received comparatively less attention when compared to UAV-assisted SAR, particularly in terms of integrated victim detection and localization. In the following, we first examine RF-based localization techniques, and then, we focus on the state of the art in Autonomous Mobile Robot (AMR)-based SAR.

A. RADIO FREQUENCY (RF)-BASED LOCALIZATION

Accurate and real-time location knowledge is a key requirement for various applications, including emergency services [20]. The latter emphasizes that radio-based localization and sensing are gaining increasing importance in 5G wireless communication systems. RF-based localization techniques leverage physical-layer signal characteristics, such as Time-of-Flight (ToF), Time of Arrival (ToA), AoA, and Direction of Arrival (DoA) to estimate the distance and direction of a target [21]. Precise estimation of these parameters is fundamental for enhancing localization precision and enabling geometric localization approaches such as trilateration and triangulation [22]. AoA and DoA can be estimated without requiring cooperation between the transmitter and receiver, enabling passive localization of targets [23]. Subspace-based algorithms such as MUSIC [24] and ESPRIT [25] have long been employed for AoA estimation, whereas recent deep learning approaches have been proposed both to enhance these traditional methods [26], [27] and to serve as alternatives to conventional techniques [28], [29], [30], [31].

Higher frequency bands, such as mmWave, offer advantages for localization through enhanced temporal and spatial resolution, which contribute to improved accuracy in complex environments, both outdoor [7] and indoor [32], [33]. In this context, a notable class of approaches leverage the directionality of the transmissions to perform localization. For example, active beamforming has been shown to significantly reduce UE localization error in mmWave Multiple-Input Multiple-Output (MIMO) systems [34], both in single-anchor scenarios [35] and in multi-anchor deployments, which further improve robustness against hardware impairments and synchronization errors [36]. Beyond conventional beamforming, Reconfigurable Intelligent Surface (RIS) technology has emerged as a promising tool for enhancing indoor localization accuracy, particularly in near-field scenarios at 28 GHz [37], [38]. RISs can support MIMO systems in jointly estimating user and scatterer positions under multipath propagation [39], and have been investigated for ToA- and AoA-based localization [40], [41]. Recent works integrate RISs with ML and RL for improved AoA estimation and target tracking [42], [43], while others report significant gains in position and orientation accuracy using particle filtering and CRLB analysis [44]. RISs have also been proposed to enable Integrated Sensing and Communication (ISAC) frameworks [45].

ISAC itself is an emerging paradigm that unifies sensing and communication within the same RF framework, improving localization efficiency and accuracy through shared signaling [46], [47], [48].

B. ROBOTICS-BASED SEARCH AND RESCUE

Focusing on AMRs for SAR, SoA studies mainly focus on mission planning solutions to enhance area coverage and minimize response time [49].

Mobile robots often rely on SLAM to explore and map unknown environments, including Light Detection and Ranging (LiDAR)-based [50], [51], graph-based [52], visual (e.g., ORB-SLAM2 [53]) and multisensor fusion [54] methods. Some works have adopted SLAM in SAR primarily for environmental mapping, where the generated maps serve as valuable tools to understand and navigate disaster sites [55], [56]. While SLAM may be a necessary condition, it is not sufficient for guiding a robot toward strategic objectives, such as reaching a detected victim located tens of meters away [4]. To address this, exploration and path planning approaches introduce frontier prioritization [57] or multi-step planning [58]. In addition to these features, explorers such as [59] propose prioritization that explicitly accounts for remote areas along adjacent ones and frontiers,

TABLE 1. Representative map coverage techniques deployed in SAR.

Ref.	Robot Type	Multirobot	Method	Prioritization	Planning
[55]	AMR	No	Exploration via ORB-SLAM	-	-
[56]	AMR	No	Exploration via 3D visual SLAM	-	-
[57]	AMR	Yes	Exploration via Rapidly-Exploring Random Tree (RRT) techniques	Frontier	Single-step
[59]	UAV, AMR	Yes	Exploration via Super Frontier Information (SFI)-driven decision-making	Global	Multi-step
[61]	Likely AMR	No	Path planning via Offline TSP	-	Multi-step
[67]	AMR	No	Exploration via Online TSP	Frontier	Multi-step
[5]	AMR	Yes	Exploration via MIP-modeled grid-based global planning	Global	Multi-step
This work	AMR	Yes	Exploration via MIP-modeled global planning with RF localization-guided prioritization	Global	Multi-step

TABLE 2. RF-enabled SAR approaches leveraging victim signals.

Reference	Scenario	Agent Type	Localization / SAR Approach	RF Technology
[14]	Outdoor	UAV	Pseudo-trilateration + ML-enhanced localization	Cellular (LTE)
[15]	Outdoor	UAV	3D hybrid positioning (ToF + AoA) + ML-based correction	5G-NR FR1 [2.5 GHz]
[16]	Outdoor	UAV	Trilateration / Multilateration + Graph Traversal (via TSP)	Generic RF signals
[6]	Indoor	UAV	RL-based trajectory optimization using RSS variation	Generic RF signals [2.4, 5.0 GHz]
[17]	Generic	UAV	Collection of RSSI from victim devices + ML-based localization	Generic RF signals
[19]	Indoor	Human	Trilateration of BLE signals	BLE [2.4 GHz]
[18]	Indoor	Human	RSSI data collection via smartphones connected to WiFi probes	WiFi
This Work	Indoor	AMR	RSSI filtering and AoA triangulation	5G mmWave [28 GHz]

while others such as [5] address both remote areas prioritization and multi-step planning by solving a Mixed-Integer Programming (MIP) problem.

Other works model the path planning problem in SAR operations as a combinatorial optimization problem such as the Traveling Salesman Problem (TSP) [16], [60], [61]. More generally, TSP-based formulations have been applied to multi-robot path planning in several variants [62], [63], [64], [65], [66].

A key distinction is between offline and online TSP approaches, depending on the initial hypothesis of the SAR operation: offline formulations assume full prior knowledge of the environment and optimize routes accordingly, while online TSP variants dynamically update the path planning as new knowledge of the environment is acquired, enabling robots to incrementally construct efficient tours in unknown environments [67], [68], [69], [70], [71].

C. CONTRIBUTIONS

Despite significant advances in indoor exploration for SAR, current approaches lack explicit integration of remote victim localization into the autonomous decision-making loop. To the best of our knowledge, this work introduces the first integrated framework that unifies exploration, remote victim localization, and operational decision-making in SAR operations with AMRs. We evaluate the proposed approach against two baselines: the explorer from [5], which adaptively plans multi-robots trajectories to optimize area coverage and mission time under resource constraints, and an online TSP-based method, e.g., [67], [69], [70], [71], that seeks to visit a set of target locations efficiently. We integrate RF-based victim localization into both the exploration loops for path replanning when new victims are detected.

Moreover, to better position our work within the SoA, we provide a comparative analysis along the two main focuses of the paper. First, in Table 1, we summarize representative exploration-based SAR methods, highlighting the type of robot, method, type of goal prioritization, and whether multi-step planning is supported. Second, in Table 2, we focus on RF-enabled SAR approaches that leverage signals emitted by victims. This table includes details on operating environment (indoor/outdoor), type of agent, RF technology, and frequency band when available.

III. PROBLEM DEFINITION: THE NEED FOR LOCALIZATION

In SAR operations involving collaborative mobile robots, it is critical to balance rapid mission completion with effective area coverage. Traditional approaches often rely on SLAM and exploration methods to navigate unknown environments. However, SAR inherently requires prioritizing rapid victim access [72]. Therefore, blindly mapping the area can be suboptimal for SAR scenarios, as it may lead robots to spend considerable time exploring empty or low-priority regions where no victims are located, thus delaying rescue efforts. To overcome this, SAR robots are typically equipped with multiple heterogeneous sensing modalities that provide complementary environmental information [1], [73]. Vision-based systems, for example, enable fast and accurate localization and detection in open or visually clear environments. Nevertheless, their performance degrades significantly in the presence of smoke, dust, or

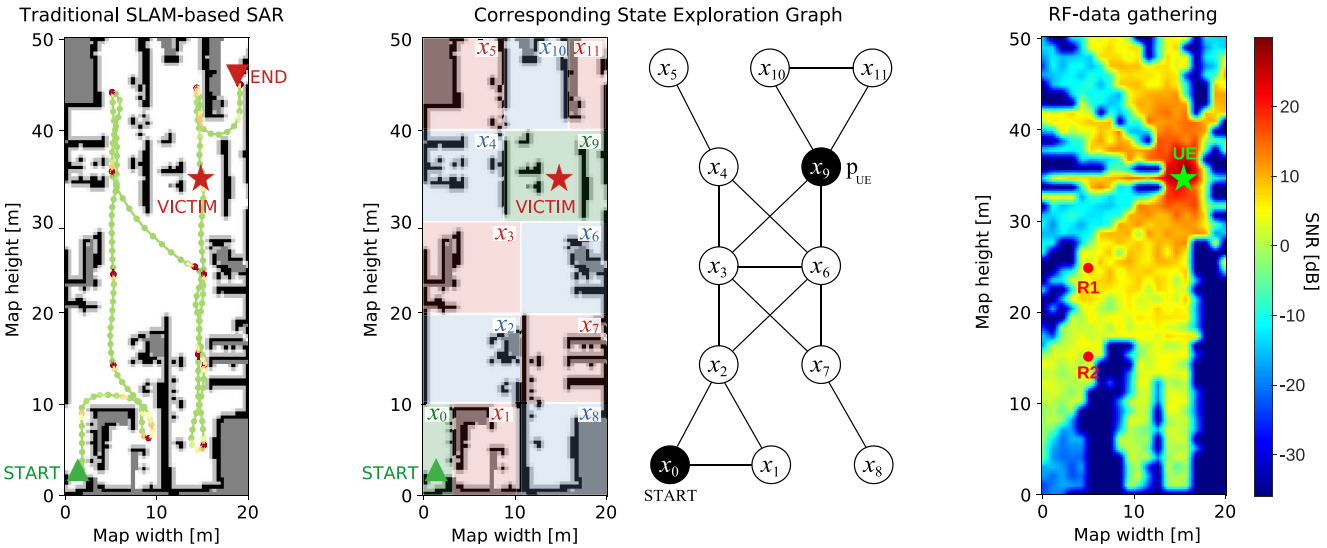


FIGURE 2. Baseline exploration trajectory for the given 20 × 50 meters indoor map; corresponding graph of exploration states; SNR heatmap for the signal sourced in the UE (victim).

low illumination, which are common in disaster sites [74]. Moreover, cameras possess limited fields of view and depend on Line-of-Sight (LOS) and orientation, making them less reliable for detecting static or partially occluded victims [75]. To overcome these limitations, we employ RF techniques to localize victims, such as AoA triangulation, augmenting classic exploration methods. The localization information obtained from RF signals can then be integrated into adaptive path planning, allowing robots to dynamically adjust their exploration strategy and prioritize regions where victims are likely to be located.

For instance, let us consider a SoA exploration method, which optimized path is shown in Fig. 2. The first plot (left-hand side) shows the optimized trajectory when deploying a single robot, according to the technique proposed in [5]. The second plot (center) presents the corresponding state-based abstraction, where each state represents a spatial region to be explored, while the third plot shows the SNR map for the RF signal sent by the UE.

Assuming that the robot starts at coordinates $(1, 2)$ in the lower-left corner of the map, corresponding to state x_0, it ascends to the top-left corner in approximately 60 seconds.[1] Subsequently, it progresses to the right half of the map, ultimately reaching the top-right corner after 215 seconds. According to this exploration strategy, a victim located in the left part of the map will be detected and reached within the first 60 seconds of the mission, allowing for a prompt intervention. However, a victim located on the right side of the map (e.g., in state x_9) will be accessed with some delay as the predefined path requires first an exhaustive exploration of the left side. This delay can compromise the effectiveness of a rescue operation by postponing victim detection.

[1]The exploration times we provide in the example are based on the simulation setup described in Section VII.

Let us consider now that the victim situated in state x_9 is carrying an active User Equipment (UE), and the possibility of localizing it through AoAs triangulation. The SNR heatmap in the third subplot of the figure under analysis illustrates that it would be possible to collect enough information to estimate the UE position earlier at state x_3. Specifically, a single robot would be able to detect the UE and perform an initial AoA estimation already in state x_2. Then, after moving to state x_3, it could perform a second AoA estimation, which can be triangulated with the first AoA to determine the position of the UE. In this example, with the aid of a single robot, leveraging RF signals enables victim detection and a first AoA estimation within 35 seconds only. Incorporating this information into the path planning process allows the robot to dynamically adjust its trajectory, reaching the victim within 60 seconds. By postponing the exploration of states x_4 to x_8, the robot significantly reduces the time required to access the victim compared to a static, predefined exploration strategy. Moreover, deploying more robots would often lead to a faster victim localization, collecting multiple AoA estimations at the same time in different positions.

It is important to emphasize that RF-based localization is intended to complement traditional sensing modalities in SAR, rather than replace them. In this work, we focus on a 28 GHz mmWave implementation, as illustrated by the SNR map of the example and deepened later in Section VI. However, the proposed approach is general and can be applied across different frequency bands, e.g., with mmWave and sub-6 GHz signals potentially used jointly to take advantage of their complementary characteristics. Indeed, lower-frequency transmissions penetrate walls and obstacles more effectively, providing more frequent but less precise localization estimates [76]. Note that the penetration ability of lower frequency signals may occasionally guide the robot toward obstacles and walls, leading to blocked exploration

Algorithm 1: AoA-Based Localization

Input : Set of received signals $\mathcal{S}$, new received signal s_{new}
Output : Estimated UE position $\hat{p}_{ue}$
Procedure: Localization

$\mathcal{S} \leftarrow \mathcal{S} \cup \{s_{\text{new}}\}$
$\mathcal{S} \leftarrow \{s \in \mathcal{S} \mid P(s) \geq min_received_power\}$
if $|\mathcal{S}| \geq 2$ **then**
  $\mathcal{S} \leftarrow \text{Sort}(\mathcal{S}, s.rx_power, \text{decreasing})$
  foreach $(s_i, s_j) \in \mathcal{S}, i \neq j$ **do**
    $\theta_i \leftarrow s_i.AoA$
    $\theta_j \leftarrow s_j.AoA$
    $\alpha \leftarrow |\theta_i - \theta_j|$
    if $\alpha \in angles_range$ **then**
      Compute_Intersection(θ_i, θ_j)

Algorithm 2: Compute PDFs Intersection

Input : AoA θ_i and θ_j
Output : Estimated UE position $\hat{p}_{ue}$
Procedure: PDFs product

$PDF_1 \leftarrow \text{ComputePDF}(\theta_i, s_i.rx_power)$
$PDF_2 \leftarrow \text{ComputePDF}(\theta_j, s_j.rx_power)$
$Probability_Area \leftarrow \text{PDFs_product}(PDF_1, PDF_2)$

paths. In contrast, mmWave-based methods provide high-accuracy localization in LOS conditions. Although their strong sensitivity to blockage and reflections in Non-Line-of-Sight (NLOS) is a communication challenge [77], it can implicitly reveal the environmental structure [78]. Indeed, the detection of reflected mmWave signals can provide valuable semantic cues about the existence of traversable paths, helping the robot plan alternative routes to reach a victim without risking ending up in blind spots or inaccessible regions from which a lower-frequency signal may still be received.

IV. VICTIM LOCALIZATION METHOD

In this section, we present the proposed localization method designed to estimate the position of a static UE by leveraging AoA measurements collected at different states of a robot's trajectory. This method utilizes geometric triangulation to determine the estimated position based on the intersection of AoA measurements of received signals.

The process requires at least two AoA measurements collected from sufficiently diverse positions. These measurements can be obtained either simultaneously, such as when multiple robots are deployed, or at different times, as in the case of a single robot that moves to collect additional data. Since the UE is assumed to be static, past measurements remain valid and can be used in conjunction with more recent ones to improve localization accuracy. Note that the assumption of a static victim is adopted to simplify the problem formulation and analysis. However, the proposed approach is not inherently limited to static scenarios: the same geometric principles underlying AoA triangulation can be extended to moving targets by processing time-sequenced angle measurements with motion models (e.g., Kalman filters [79] and particle filtering [80]).

As formalized in Algorithm 1, upon receiving a new signal s_{new}, it is added to the set of previously received signals $\mathcal{S}$ and the corresponding RSSI is recorded as an indicator of the received power. To ensure reliable localization, the algorithm filters out all signals with a received power below the threshold *"min_received_power"* retaining only strong signals that are less likely to be affected by reflections and noise. If fewer than two signals remain after filtering, the algorithm terminates, as localization is not feasible.

Next, the algorithm sorts the signals in $\mathcal{S}$ in decreasing order of received power, prioritizing the strongest ones, and iterates over pairs of signals (s_i, s_j). A pair is considered suitable for triangulation if the angle formed by their AoA measurements $\alpha = |\theta_1 - \theta_2|$ falls within a range that ensures a geometrically reliable intersection. Specifically, the intersection angle must be neither too small, where lines are nearly parallel, nor too large, where lines are nearly antiparallel. If no signal pair satisfies this criterion, the algorithm concludes that a reliable localization is not possible at that time. Conversely, if a valid pair is found, the intersection of their AoA directions is computed and used to estimate the UE position.

A. PROPOSED APPROACHES

To adapt the localization method to different computational constraints and application requirements, we propose two implementations for the AoA-based intersection step: a probabilistic approach and a deterministic alternative. These two variants offer a trade-off between modeling accuracy and computational efficiency.

1) PROBABILISTIC IMPLEMENTATION

In the probabilistic implementation (Algorithm 2, Fig. 3), each AoA is modeled as a Gaussian Probability Density Function (PDF) centered around the measured angle, according to [42], [81], [82]. The estimated UE position is obtained by computing the product of the two spatial likelihoods derived from the AoA distributions. This results in a 2D probability map where the overlapping high-probability regions suggest the likely position of the UE.

The standard deviation of each AoA's distribution is adapted based on the received signal's SNR, following an exponential decay relationship as suggested in [83] and [84].[2] This reflects the intuition that weaker or noisier signals introduce greater angular uncertainty, resulting in broader likelihood distributions. This method (Algorithm 2) is based on explicit uncertainty quantification but incurs a complexity of $\mathcal{O}(N^2)$ due to the discretized grid-based evaluation of the 2D probability map, where N is the number of cells per axis

[2]The model we adopt to link AoA distribution and SNR is detailed in Section VI.

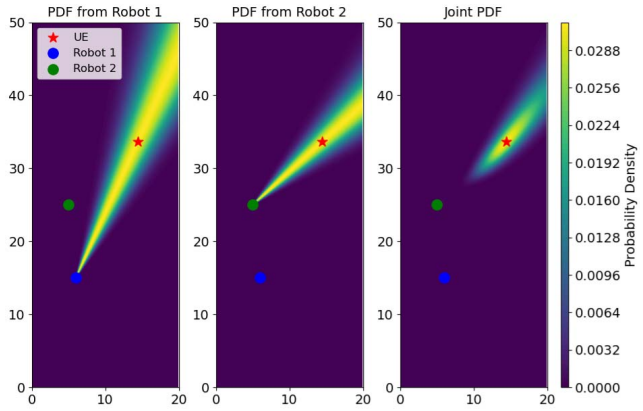


FIGURE 3. AoAs modeling via PDFs. The UE estimated position falls in the resulting section of probability area.

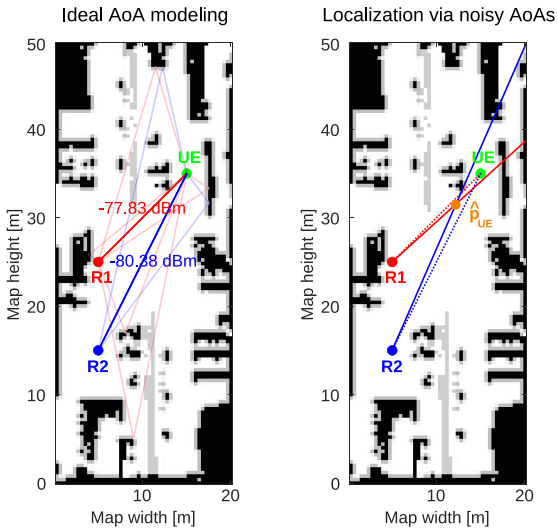


FIGURE 4. UE Localization via deterministic AoA triangulation. The methodology can be performed in one step by deploying two robots, or alternatively in two steps with one robot that collects two AoA in two different positions.

used to discretize the 2D space (e.g., 200 × 500 cells when studying a 20 × 50 meters map discretizes with a resolution of 0.1 meters per cell). This includes computing the product of the PDFs at each cell and normalizing the resulting map.

2) DETERMINISTIC IMPLEMENTATION

The deterministic implementation (Fig. 4) offers a lightweight alternative by treating each AoA as a noisy but fixed direction. Rather than evaluating the intersection of full probability maps, this method geometrically intersects the two semilines, producing an estimated position $\hat{p}_{ue}$ by solving a 2 × 2 linear system. This approach significantly reduces computational cost, achieving constant-time complexity $\mathcal{O}(1)$, making it suitable for real-time and embedded scenarios.

V. SYSTEM ARCHITECTURE

Classical exploration methods are typically structured around either a two-layered architecture composed of a *Global Planner* and a *Local Planner*, or a single centralized planner. In the first case, the separation of the two-layered approach ensures scalability and responsiveness: the global planner manages long-term, strategic decision-making over large environments, while the local planner handles real-time navigation and obstacle avoidance based on immediate sensor data. In contrast, a single centralized planner integrates both global and local decision-making within a unified framework, yielding highly coordinated motion plans at the cost of increased computational complexity and reduced scalability.

In this work, we propose to enhance the path planning process by integrating a novel input: the estimated localization of remote targets. This new input enables the planner to reason not only about unexplored space, but also about reachable, higher-priority locations. To implement this integration, we adopt existing exploration frameworks as baselines: *i*) the SoA approach proposed in [5], which employs a two-layer hierarchical planning architecture; and *ii*) a TSP-based exploration strategy approach employing a greedy heuristic (e.g., [69]), which follows a single-planner formulation. The two selected baseline approaches are detailed as follows.

A. HIERARCHICAL EXPLORATION ARCHITECTURE

In the hierarchical planning framework, multi-step planning and area prioritization are handled at the global level through periodic planning cycles, while the local layer determines how to explore the selected regions with continuous, context-aware adjustments. The **Global Planner** adopts a grid-based exploration strategy based on the discretization of the environment into a set of regions. It maintains a tabular representation of explored and unexplored areas, where each entry encodes the area's exploration status, environmental constraints (i.e., traversal costs to adjacent areas), and assigned priority. The planner formulates the goal assignment process as a Mixed-Integer Programming (MIP) problem,[3] enabling coordinated decision-making across multiple robots and avoiding conflicts: binary decision variables represent robot-to-area assignments, and constraints enforce that each area is assigned to at most one robot, unless the area is part of a shared-access region (e.g., bottlenecks or narrow passages that all robots must traverse to reach unexplored regions). It computes global goals by optimizing over a planning window W, which defines the number of areas considered during each planning cycle. Then it updates the exploration strategy when a new area is explored, enabling strategic exploration based on current knowledge.

The **Local Planner** refines global goals using LiDAR-based SLAM. It continuously processes sensor data to set precise local objectives, avoid dynamic obstacles, and overcome occlusions. This layer also supports real-time map merging to coordinate decisions among multiple robots.

Specifically, RF-based information is collected and transmitted to the Global Planner whenever a robot explores a scheduled area. This enables the latter to incorporate

[3]More details about the Problem Formulation can be found in [5].

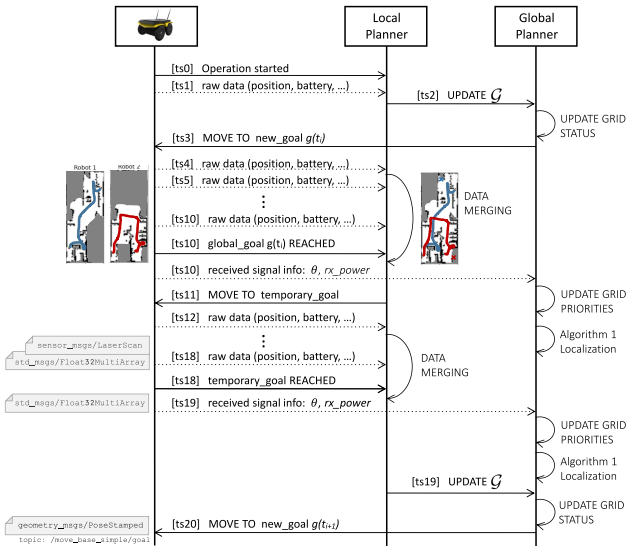


FIGURE 5. The sequence diagram highlights the exchange of messages among the three layers within a time frame reserved to explore the current area. Examples of ROS messages are shown with grey labels. The frequency of occurrence is emulated with example timestamps *ts*. Dashed lines are sensors data messages, while continuous lines are commands and status updates.

probabilistic victim cues into its periodic updates and identify the area to prioritize, refining the long-term exploration strategy with task-relevant information. Both planners are designed to coordinate the robots in a centralized way, maintaining aggregated representations built from the data streams of all active robots.

To clarify the system's operational dynamics, Fig. 5 illustrates the data and command flows between components through a sequence diagram. Dashed arrows represent raw sensor data transmissions (e.g., LiDAR and RF signals information), while solid arrows denote high-level commands or status updates. Example timestamps are included to emulate the typical frequency of each interaction and convey the asynchronous nature of the system. More details about messages implementation are disclosed in Sections VI-A.

LiDAR data is continuously collected and transmitted by individual robots to the Local Planner, which aggregates it to a centralized costmap. As a result, the Robot and Local Planner layers operate in a continuous manner, while the Global Planner follows an event-driven strategy: it updates the abstract grid representation status and assigns exploration priorities when new localization cues or goal completion updates are received.

B. TSP-BASED EXPLORATION ARCHITECTURE

The Traveling Salesman Problem (TSP) is a combinatorial optimization problem where, given a set of locations and pairwise travel costs, the objective is to find the shortest possible tour that visits each location exactly once and returns to the starting point.

When operating on occupancy grid maps generated from 2D LiDAR data, the problem can be cast into Traveling Salesman Problem (TSP) form by first extracting relevant waypoints, such as the centroids of frontiers. The set of targets evolves dynamically as the robot discovers new areas of free space, which may generate additional frontiers. The TSP tour is therefore recomputed from the robot's current position, ensuring that planning accounts for the most recent map information. This formulation is referred to as the *"online TSP"*, in contrast with the *"offline TSP"*, where the full map is assumed to be known in advance and the tour can be computed once before deployment. Given the NP-hard nature of the problem, we implement a greedy nearest-frontier strategy in the online setting, a commonly used baseline in prior works [69], [70], [71]. This approach serves as our TSP-based method, while the localization-aware variant extends this strategy by replanning upon the detection of victims, replacing the standard frontier selection to prioritize victim access with Dijkstra's algorithm.

The complete procedure is reported in Algorithm 3, and the main steps are as follows:

1) The set of frontier nodes F visible from the current robot position p is defined (line 1);
2) A weighted graph G connecting the current position p with all frontier nodes is computed using travel costs as edge weights (line 3);
3) The next frontier node n is chosen according to the greedy strategy (line 4);
4) Once the robot reaches the target frontier point (line 5), the set of frontiers is updated (line 7), and new RF data is gathered as in Algorithm 1.

The TSP terminates either when no frontiers are left, indicating that the environment has been fully explored, or when the target has been localized. In this second case, the exploration continues iteratively prioritizing access to the victim using Dijkstra's algorithm (lines 12 and 13).

VI. SCENARIO SETUP

A. ROBOTICS PHYSICS-BASED SIMULATION

1) SIMULATION ENVIRONMENT

The experimental evaluation is conducted using the ROS framework [85] and the Gazebo simulator [86],[4] commonly used in the robotics research community to validate algorithms before real-world deployment [87], [88].

The planning architecture is deployed on edge nodes that communicate via ROS messages, enabling coordination and control of a fleet of AMRs. Despite the distributed implementation, the decision-making process remains centralized: the planners determine robot behaviors and navigation goals without requiring any autonomous decision-making at the robot level.

Specifically, the Local Planner is implemented on a remote server as a C++ ROS node that processes each robot's 2D LiDAR data and issues new local goal commands. Communication occurs via the publisher-subscriber paradigm typical of ROS, where nodes publish messages on dedicated

[4]https://gazebosim.org/home

Algorithm 3: Localization-Aware Online TSP

Input : Occupancy grid map M,
initial position p_0

Initialize : $p \leftarrow p_0$;
$\hat{p}_{ue} \leftarrow null$;
$target_reached \leftarrow$ **false**;
$F \leftarrow \emptyset$; // Frontier set

Procedure:

$F \leftarrow$ Detect_Frontiers (M, p) ;
while $F \neq \emptyset$ *and* $\hat{p}_{ue} = null$ **do**
  $G \leftarrow$ Compute_TSP_Graph (F, p) ;
  $n \leftarrow$ Select_Next_Node (G) ;
  $p \leftarrow$ Move_To (n) ;
  $\hat{p}_{ue} \leftarrow$ AoA_Localization (M, p) [Alg.1];
  $F \leftarrow$ Detect_Frontiers (M, p) ;
while $target_reached =$ **false do**
  $F \leftarrow$ Detect_Frontiers (M, p) ;
  if $F = \emptyset$ **then**
    break ;
  $n^* \leftarrow$ Select_Best_Node (F, $\hat{p}_{ue}$) ;
  $p \leftarrow$ Move_To (n^*) ;
  if $\hat{p}_{ue} \in$ *Lidar_View (p)* **then**
    $target_reached \leftarrow$ **true** ;

topics, and other nodes subscribe to those topics to access data. Examples of ROS messages we use are reported in Figure 5, namely: *sensor_msgs/LaserScan* for LiDAR data; *std_msgs/Float32MultiArray* to communicate multiple scalar data such as current robots' pose and battery levels; *geometry_msg/PoseStamped* to control the navigation of the robot.

The Global Planner is implemented in SIMULINK and MATLAB on a second remote server. It subscribes to the topics providing information on local area exploration progress and robots' state and battery levels. It publishes global navigation targets using separate namespaces for each robot. The goal assignment is formulated as a MIP problem and solved using the Gurobi Optimizer,[5] integrated through its official MATLAB interface. This planner interfaces directly with a MATLAB-based ray tracer that computes the RSSI values based on the robots' positions and orientations. In a real-world system, these measurements would typically be acquired via onboard RF modules and transmitted over a ROS topic to which the planner would subscribe. A high-level overview of the system architecture is presented in Fig. 1. The same reasoning and setup apply to the TSP-based planner, which is implemented on a remote server that interfaces with the ray tracer and directly with the robot plane, without passing through intermediate levels, as opposed to the previous case.

[5]https://www.gurobi.com

We deploy between 1 and 3 Jackal wheeled robots,[6] each equipped with an 8-meter radius 2D LiDAR, to explore the 20 × 50 meters realistic indoor environment chosen to test the baseline framework. This is a custom map created from the open-source factory,[7] and contains various obstacles, including shelves, stations, and two conveyor belts.

The latter effectively serve as barriers for the robots (higher cost in the costmap) while not compromising RF signals transmission. All the other obstacles reflect RF signals up to a single indirect bounce (i.e., one hop), as described in the next section. Their disposition and semantics are shown in Fig. 4, which overlays 2D LiDAR points of the explored map. In this representation, black pixels denote obstacles that block both robot movement and RF signal propagation, whereas gray pixels represent structures, such as conveyor belts, that hinder robot motion but are partially transparent to RF signals, since only their legs cause obstruction.

2) SCALABILITY CONSIDERATIONS

The decision to deploy up to three robots is based on prior analysis within the baseline framework and ensures a consistent comparison, which demonstrated that this team size is sufficient for exploring medium-sized indoor environments of similar complexity. Adding a fourth robot would be redundant and would not lead to a notable improvement in exploration efficiency. In general, environments with a larger area may benefit from deploying additional robots. The scalability of the framework is supported by the modular design of both the global and local planners: the first is tested to handle grids that abstract up to 200 × 200 meters environments, while the second computes N local views in parallel, one per robot, without significantly impacting per-robot performance. However, a key limiting factor in scalability lies in the planning horizon, defined by the window W of potential next steps (e.g., $W = 10$), regardless of environment size.

3) VICTIM DISPOSITION

To analyze localization performances in indoor environments, we study the signal characteristics based on the AoAs from a UE. Our goal is to comprehensively cover the map to account for both easy and challenging scenarios, as well as outliers (e.g., UEs that are completely hidden and whose signals are not received). Specifically, we consider 500 positions for the UEs, which are randomly distributed within the 20 × 50 meters map to ensure broad coverage. The UEs are placed on valid positions by analyzing the 2D LiDAR-based occupancy map, where white pixels represent areas suitable for placement while dark pixels indicate obstacles.

[6]https://clearpathrobotics.com/jackal-small-unmanned-ground-vehicle/
[7]https://github.com/mlherd/Dataset-of-Gazebo-Worlds-Models-and-Maps/tree/master/worlds/factory

B. RAYTRACING DETAILS

1) WORKING FREQUENCY AND CHANNEL MODELING

In our study, we designed a simulation environment operating at a carrier frequency of 28 GHz, belonging to the mmWave spectrum, which is known for its high temporal resolution, strong spatial directivity, and limited multipath effects [7]. Characteristics that are beneficial for precise localization. The choice of frequency also aligns with recent empirical studies conducted in comparable indoor industrial environments with dimensions close to our test area [89].

The propagation environment was modeled using a 2D ray-tracing framework developed in MATLAB. The choice of a 2D custom ray-tracer, instead of more sophisticated 3D models, is motivated by the need for seamless integration within a SAR robotics-related loop, while still providing sufficient details for indoor navigation [90]. Our setup follows the configuration guidelines provided in [91] and integrates parameters from the standardized 3GPP channel model [92]. Within this framework, both LOS and NLOS conditions are explicitly considered, with path loss characterized according to the following equations:

LOS Path Loss:

$$PL_{\text{InH-LOS}} = 32.4 + 17.3\log_{10}(d_{3D}) + 20\log_{10}(f_c), \quad (1)$$

with a shadow fading standard deviation of $\sigma_{SF} = 3$ dB.

NLOS Path Loss:

$$PL_{\text{InH-NLOS}} = \max\left(PL_{\text{InH-LOS}}, PL'_{\text{InH-NLOS}}\right), \quad (2)$$

where

$$PL'_{\text{InH-NLOS}} = 38.3\log_{10}(d_{3D}) + 17.30 + 24.9\log_{10}(f_c) \quad (3)$$

with shadow fading deviations of $\sigma_{SF} = 8.03$ dB. Both LOS and NLOS models are valid for $1m \leq d_{3D} \leq 150m$, where d_{3D} corresponds to the Euclidean distance between transmitter and receiver in LOS, whereas it corresponds to the length of the reflected path in NLOS. Reflections are modeled as specular when the signal encounters propagation obstacles such as walls. These are treated as opaque bodies with reflective boundaries, and their edges are assumed to act as reflecting surfaces. The position and size of obstacles are determined from the costmap of the 20 × 50m area under study. Since we investigate a 2D scenario, ground and ceiling reflections are not explicitly traced but are statistically included in the propagation model through random fading. NLOS paths are considered up to the first reflection (single-hop), as higher-order reflections at mmWave frequencies are generally negligible due to severe power loss [93].

2) TRANSMITTER AND RECEIVER DETAILS

The transmitter, representing the UE, considered the victim in the context of SAR scenarios, is modeled as an omnidirectional radiator with an Effective Isotropic Radiated Power (EIRP) of 24 dBm, in accordance with the specification in [94].

The receiver is abstracted as an antenna array with directional reception capabilities, thereby enabling AoA estimation. To maintain generality and applicability across diverse receiver implementations, we do not commit to a specific array geometry, size, or AoA estimation algorithm. Instead, we model the AoA estimation accuracy through an error distribution. Further details on the array assumptions, estimation process, and uncertainty characterization are provided later in this section. Both the direct LOS and the reflected NLOS signal components are considered at the receiver side. Among them, the most powerful component is selected as the primary signal. This ensures that the localization process is based on the strongest and most reliable path available, minimizing errors caused by weaker multipath reflections.

For the NLOS case, the model considers a single reflection (one-hop propagation), discarding multipath components with two or more bounces due to excessive power loss. Moreover, all the signals characterized by a received power below the *min_received_power* threshold set to −85 dBm are discarded to ensure reliable position estimation (Algorithm 1, line 2). This threshold guarantees a SNR within the range of 0 to 10 dB in the worst-case scenario of our experiments, where localization remains feasible and aligns with *robust operation* in low-SNR conditions (e.g., −5 dB to 5 dB), particularly when employing beamforming or subspace-based techniques such as MUSIC [95], [96].

Specifically, the SNR is defined as the difference between the received signal power and the noise power, both measured in dBm. Considering a noise power of approximately −87 dBm,[8] corresponding to a working frequency of 28 GHz and a receiver bandwidth of around 100 MHz, the threshold for the minimum received power set to the −85 dBm ensures operation within the intended low-SNR regime even in worst-case scenarios.

Lastly, the AoA of each received signal is modeled as a Gaussian PDF, centered around the simulated AoA with a standard deviation that depends exponentially on the received power and SNR [83], [84]. Specifically, these works demonstrate empirically that the AoA estimation error decreases exponentially with increasing SNR. We adopt the following model for the standard deviation:

$$\sigma(SNR) = a \cdot e^{-b(SNR)} + \epsilon \quad (4)$$

where the parameters used to fit are an amplitude $a = 0.075$, a decay rate $b = 0.1$, and the lower bound error $\epsilon = 0.05$. These yield a standard deviation of 0.125 radians (∼ 7 degrees) when $SNR = 0$ dB, tending asymptotically to $\epsilon = 0.05$ radians (∼ 3 degrees) for the strongest signals received near the source. These standard deviation values reflect a deliberately conservative choice, slightly overestimating the angular error compared to the bounds reported in [101], in order to avoid overestimating system performance. Specifically, the latter reports values

[8] We consider a thermal noise floor of −174 dBm/Hz over 100 MHz and a noise figure up to 7 dB for 28 GHz implementations as per [97], [98], [99], [100].

of Orientation Error Bound (OEB) spanning from less than 1 degree to 10^{-2} degrees for different numbers of antennas (4 to 256), comparing different methods working at a frequency of 28 GHz. This conservative approach abstracts away the specifics of the receiver implementation: a receiver with as few as 4 antennas can achieve performance comparable to or exceeding the values reported here, independently of the specific AoA estimation algorithm used (e.g., MUSIC, ESPRIT, or other standard methods), while additional antennas further improve localization accuracy. At the same time, this approach models a range of non-ideal conditions, including hardware imperfections and external interference, ensuring that our reported results represent a lower bound on the performance achievable in practice.

Eq. (4) is also applied in the deterministic mode to model the uncertainty in the AoA measurements. Here, we repeat the triangulation process multiple times (e.g., 100 iterations for each experiment) to avoid relying on potentially favorable or anomalous outcomes. In each iteration, the AoA is perturbed by a noise term $\eta \sim \mathcal{N}(0, \sigma)$ based on the signal quality.

VII. PERFORMANCE EVALUATION

A. METRICS DESCRIPTION AND MOTIVATION

In this section, we present the metrics used to evaluate the performance of our localization model and explain the motivations behind their selection. Specifically, we employ the Cramér-Rao Lower Bound (CRLB) and the Root Mean Square Error (RMSE) for the probabilistic model, and the Euclidean distance for the deterministic approach.

1) CRAMÉR–RAO LOWER BOUND

For the probabilistic model, the target's position is estimated by combining the information from probabilistic AoA measurements. Each AoA measurement is modeled as a PDF, and the intersection of multiple such PDFs forms a probability region that characterizes the spatial likelihood of the target's presence. The area of this region reflects the precision of the localization process: smaller areas typically indicate higher confidence and lower uncertainty. This concept is closely related to the CRLB, which provides a theoretical lower bound on the variance of any unbiased estimator, as derived from the Fisher Information Matrix (FIM).

In a 2D localization scenario, the FIM is defined as [102]:

$$\mathbf{F} = \begin{bmatrix} F_{xx} & F_{xy} \\ F_{yx} & F_{yy} \end{bmatrix} \tag{5}$$

where each element is given by the expected value of the product of partial derivatives of the log-probability density function $Z(\mathbf{x} \mid x, y)$ with respect to the parameters:

$$F_{xx} = \mathbb{E}_{\mathbf{x}|x,y}\left[\left(\frac{\partial}{\partial x}\ln Z(\mathbf{x} \mid x, y)\right)^2\right] \tag{6}$$

$$F_{yy} = \mathbb{E}_{\mathbf{x}|x,y}\left[\left(\frac{\partial}{\partial y}\ln Z(\mathbf{x} \mid x, y)\right)^2\right] \tag{7}$$

$$F_{xy} = F_{yx} = \mathbb{E}_{\mathbf{x}|x,y}\left[\frac{\partial}{\partial x}\ln Z(\mathbf{x} \mid x, y)\ \frac{\partial}{\partial y}\ln Z(\mathbf{x} \mid x, y)\right] \tag{8}$$

In practice, we approximate them using finite differences. The partial derivatives are computed as:

$$\frac{\partial}{\partial x}\ln Z(\mathbf{x} \mid x, y) \approx \frac{\ln Z(\mathbf{x} \mid x + \Delta x, y) - \ln Z(\mathbf{x} \mid x, y)}{\Delta x} \tag{9}$$

$$\frac{\partial}{\partial y}\ln Z(\mathbf{x} \mid x, y) \approx \frac{\ln Z(\mathbf{x} \mid x, y + \Delta y) - \ln Z(\mathbf{x} \mid x, y)}{\Delta y} \tag{10}$$

Substituting these into Eqs. (6)–(8) gives the finite-difference approximations:

$$F_{xx} \approx \mathbb{E}_{\mathbf{x}|x,y}\left[\left(\frac{\ln Z(\mathbf{x} \mid x + \Delta x, y) - \ln Z(\mathbf{x} \mid x, y)}{\Delta x}\right)^2\right] \tag{11}$$

$$F_{yy} \approx \mathbb{E}_{\mathbf{x}|x,y}\left[\left(\frac{\ln Z(\mathbf{x} \mid x, y + \Delta y) - \ln Z(\mathbf{x} \mid x, y)}{\Delta y}\right)^2\right] \tag{12}$$

$$F_{xy} \approx \mathbb{E}_{\mathbf{x}|x,y}\left[\left(\frac{\ln Z(\mathbf{x} \mid x + \Delta x, y) - \ln Z(\mathbf{x} \mid x, y)}{\Delta x}\right) \cdot \left(\frac{\ln Z(\mathbf{x} \mid x, y + \Delta y) - \ln Z(\mathbf{x} \mid x, y)}{\Delta y}\right)\right] \tag{13}$$

The FIM provides a lower bound on the covariance of any unbiased estimator, known as the CRLB. Specifically, under regularity conditions, the CRLB states that the covariance matrix of any unbiased estimator $(\hat{x}, \hat{y})$ satisfies:

$$\mathrm{Cov}(\hat{x}, \hat{y}) \succeq \mathbf{F}^{-1} \tag{14}$$

meaning that the inverse of the FIM, $\mathbf{F}^{-1}$, represents the minimum achievable covariance for any unbiased estimator of the parameters. Therefore, $\mathbf{F}^{-1}$ is referred to as the CRLB.

A common scalar measure of this uncertainty is the so-called CRLB, which is defined as the square root of the trace of the inverse of the FIM:

$$\sigma_{\mathrm{CRLB}} = \sqrt{\mathrm{Tr}(\mathbf{F}^{-1})} \tag{15}$$

where $\mathrm{Tr}(\cdot)$ denotes the trace of the matrix, which captures the minimal expected total standard deviation in position estimation.

2) ROOT MEAN SQUARE ERROR

Precision alone might not fully describe the quality of the estimation, especially in environments affected by multipath or NLOS conditions, where the true target position may lie outside the high-probability region. In such cases, point-based metrics or variance bounds may underestimate the actual error. To account for this, we also adopt a distribution-aware performance measure based on the RMSE integrated over the estimated position PDF, in addition to the σ_{CRLB}. Specifically, we compute the expected squared distance between every point $\mathbf{x}$ in the probability area and the true target location $\mathbf{x}_{\mathrm{true}}$, weighted by the posterior density $p(\mathbf{x})$:

$$\mathrm{RMSE}_{\mathrm{pdf}} = \sqrt{\int \|\mathbf{x} - \mathbf{x}_{\mathrm{true}}\|^2\, p(\mathbf{x})\, d\mathbf{x}} \tag{16}$$

This formulation captures not only the spread of the probability distribution but also its displacement from the

actual target location. As such, it provides a robust and comprehensive evaluation of both the precision and accuracy of AoA-based localization methods, particularly under uncertainty and model mismatch.

3) EUCLIDEAN DISTANCE

In the deterministic model, the target's position is estimated by intersecting two lines corresponding to the AoA. To evaluate the accuracy of this method, we use the Euclidean distance between the estimated position and the true target location, given by:

$$d_{\text{eucl}} = \|\hat{p}_{ue} - p_{ue}\| \tag{17}$$

For a single iteration, this metric directly measures the spatial error, providing a simple and effective evaluation of the position estimate's accuracy.

However, since noise in the deterministic model is modeled as angular shifts rather than probabilistic uncertainty, we also consider the Euclidean distance averaged over multiple iterations to better capture the performance under varying conditions, where each time the ideal AoA is shifted of a random η.

B. LOCALIZATION ACCURACY: METRICS COMPARISON

To evaluate the quality of the localization process, we assess the previously described metrics across 500 independent experiments, each involving a single static victim whose position is to be estimated. The victims are distributed randomly across the map (Section VI). Each experiment is repeated by deploying $N = \{1, 2, 3\}$ robots, which execute Algorithm 1 at each exploration state x_i. The localization attempt is performed based on the signals received up to that point, yielding a corresponding value for each evaluation metric. As a result, each experiment generates one metric value per exploration state, leading to a total of $500 \times S$ data points, where S is the number of exploration states.

The 3D scatterplots in Fig. 6 illustrate how the metrics under study vary as a function of three key input features: the power of the strongest signal collected up to state x_i, the power of the second strongest signal, and the angle of intersection between the two signals. These visualizations offer valuable insights into how environmental and geometric factors influence localization performance across time and configurations. In this analysis, no filtering is applied to the intersection angle, which in practice is typically constrained to avoid near-parallel configurations. This decision enables us to explore how extreme intersection angles (i.e., those close to 0° or 180°) impact localization error, and to empirically determine a suitable operational range for reliable angle intersections.

Specifically, the localization accuracy achieved by the probabilistic positioning method is evaluated using the σ_{CRLB} and the RMSE (Eq. (16)), respectively in the first and the second row of scatterplots (Fig. 6). The former confirms its role as a theoretical lower bound, generally mirroring the trends observed in the RMSE. Similarly, the third row in the figure illustrates the localization performance when applying the deterministic approach. Also, the averaged Euclidean distance exhibits similar trends, accentuating, however, the cases where the AoA are antiparallel. This behavior supports our initial hypothesis that the deterministic approach is less robust than the probabilistic one, especially under conditions where intersection geometry is poor and small angular shifts lead to significant errors. To enhance the interpretability of the color scale and better highlight differences among lower RMSE values, errors exceeding 15 meters have been clipped to 15. This prevents high outliers from skewing the color gradient and obscuring distinctions between more relevant accuracy classes. Moreover, these high-error points correspond to specific *angles_ranges* that are not considered in the subsequent analysis.

The observed trends are consistent with theoretical expectations and offer practical guidance for system design:

- *Intersection Geometry:* The accuracy of localization is strongly influenced by the relative angle between AoA lines. Performance degrades significantly when the received angles are parallel or antiparallel.
- *Signal Strength:* Stronger signals generally lead to lower estimation error, but the impact is often secondary to that of the AoA geometry. Notably, even moderate signal strengths, such as −85 dBm, often yield satisfactory localization performance when the intersection geometry is favorable. In many cases, such signals are received earlier in the exploration process, enabling timely localization and reducing the overall mission duration.

These empirical findings support the definition of an effective operating range for the intersection angle. Specifically, we identify $35° \leq \alpha \leq 170°$ as an acceptable AoAs intersection angle (*"angles_range"* for Algorithm 1), within which the metrics record values below 5 m for the majority of the points.

C. LOCALIZATION BENEFITS: OPERATION TIME SAVINGS FOR HIERARCHICAL EXPLORATION

To evaluate the impact of victim localization in SAR operations, we randomly place a single static victim per experiment. To ensure a comprehensive evaluation of the system's adaptability and effectiveness, we conduct multiple experiments, each with the victim positioned differently as described in the Scenario Setup (Section VI). Moreover, in addition to standard localization accuracy metrics, we introduce several time-based performance metrics to capture the practical benefits of early victim detection. These include:

- the "access time" as the time required for the baseline approach to access physically the victim via SLAM,
- the "localization time" t_{loc} as the time required to estimate the victim's position $\hat{p}_{ue}$,
- the "localization leading time" t_{lead} as how early we can sense a victim in relation to the baseline method;
- the "time saved" t_{saved} as the difference in time between reaching the victim using the updated path planning and

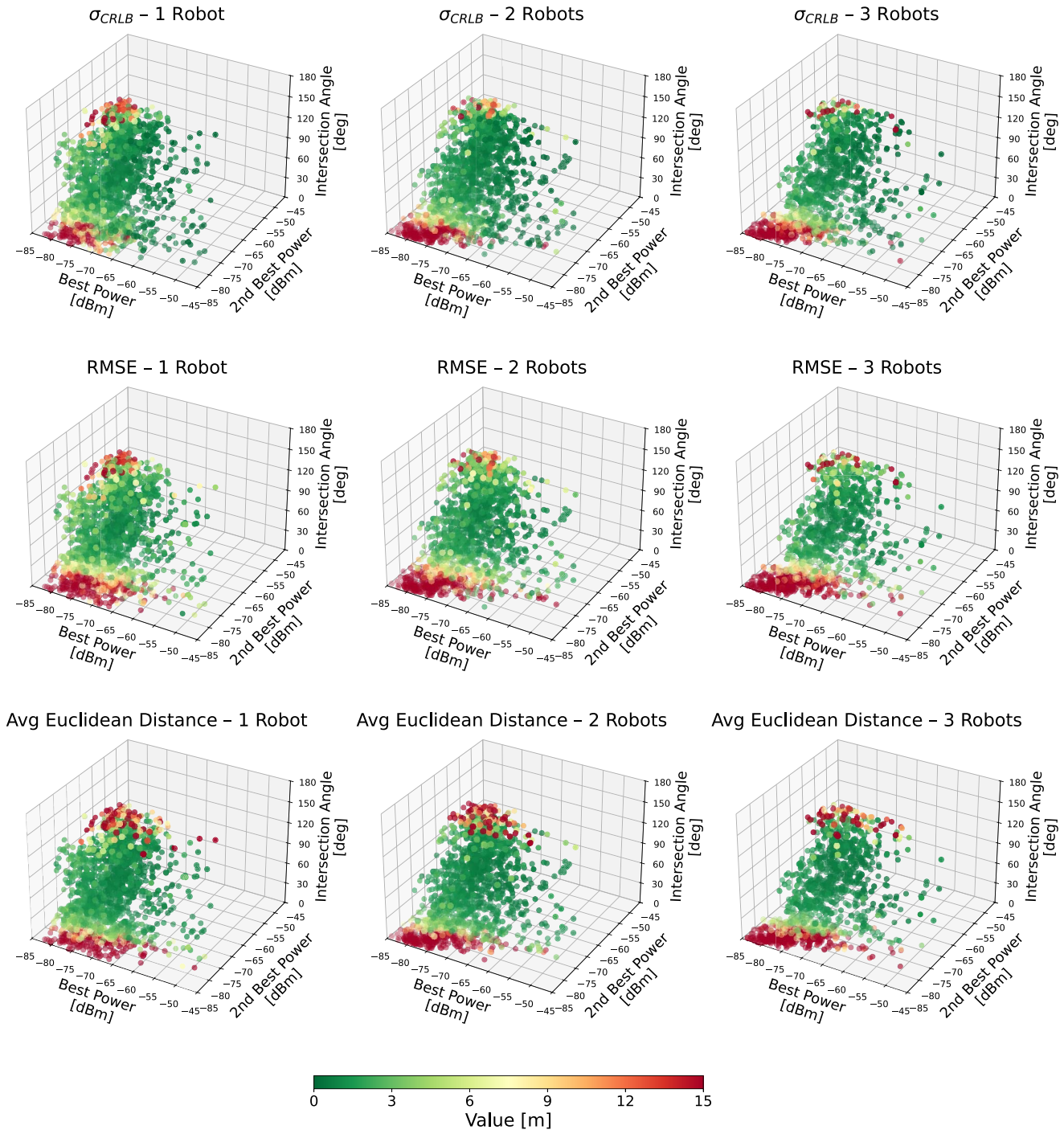


FIGURE 6. **Accuracy metrics for the Probabilistic and Deterministic Localization methods. The 3D scatter plots illustrate the localization accuracy for 500 targets across different exploration states, highlighting how the three input variables affect the output. Results are shown for deployments with $N = \{1, 2, 3\}$ robots.**

the time required following the original method without victim localization.

In the performance evaluation, we compare our method against the benchmark [5]. Figure 7 compares the time metrics (in seconds) for a single-robot deployment. The first plot shows the baseline trajectory to provide direct visual feedback, followed by the access times for each victim in the second panel. The third illustrates the *Localization Time* (t_{loc}), which corresponds to the time required to collect at least two received signals with power greater than -85 dBm, ensuring that the *AoA* intersection falls within the *angles_range* $= [35°, 170°]$.

It is important to note that the actual localization time must be adjusted to the minimum value between the defined t_{loc} and the access time, as the victim is automatically localized while being reached. For instance, in certain cases—such as

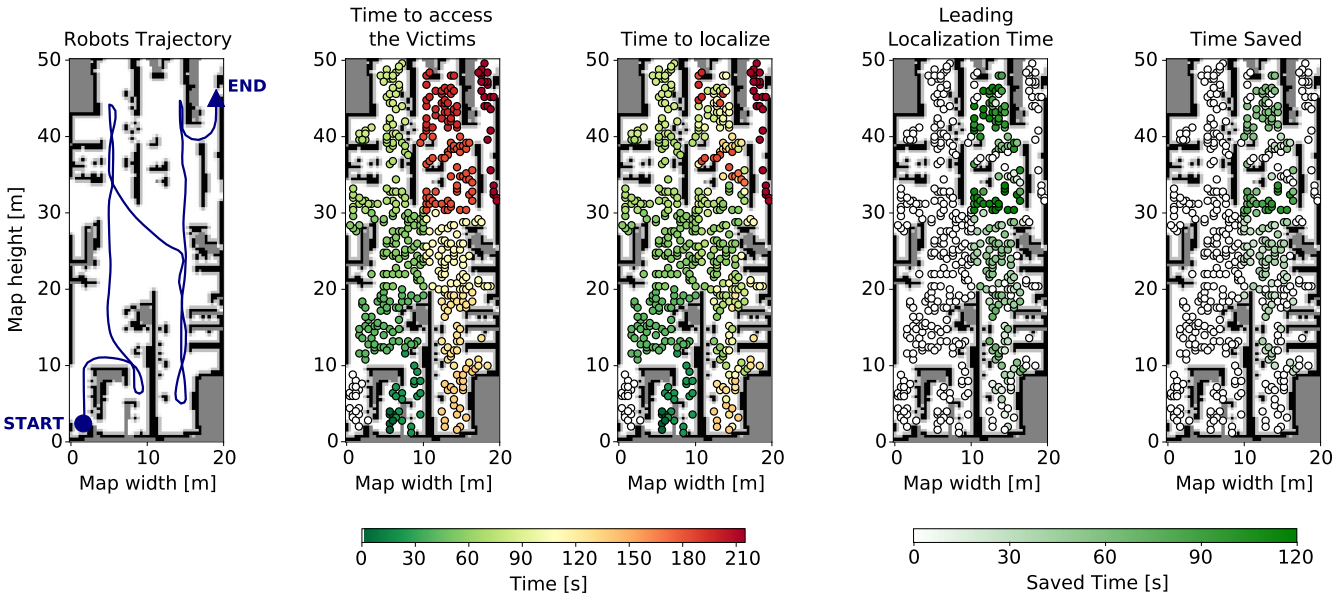


FIGURE 7. Comparison of key time metrics for a single-robot deployment, preceded by a trajectory plot for better context.

Error distributions along different intersection angle ranges (1 robot):

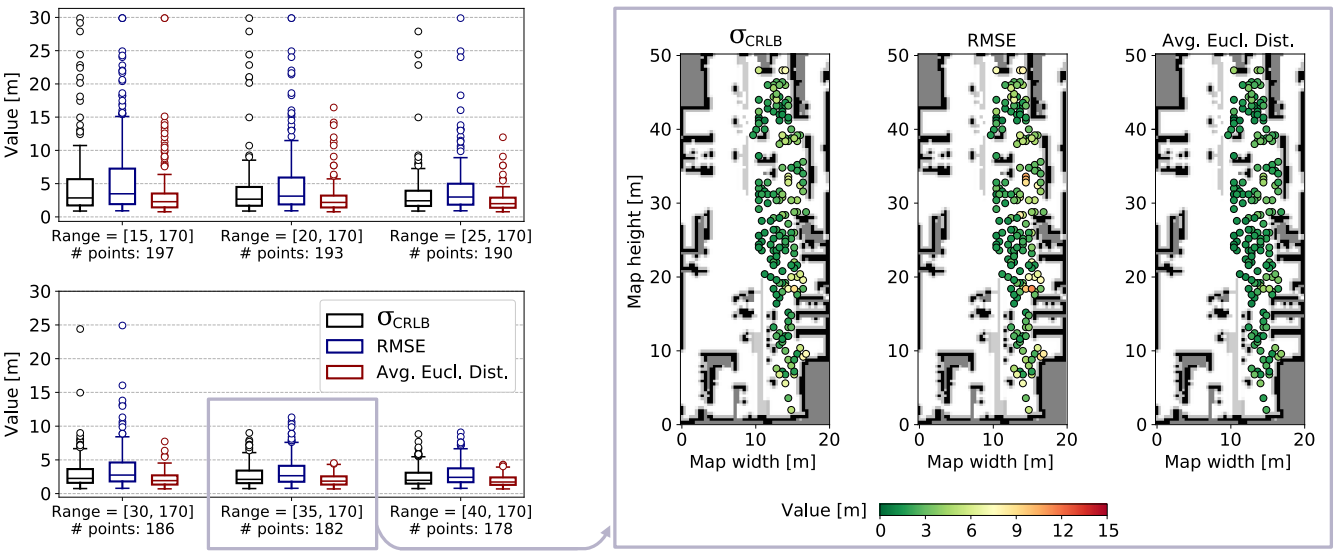


FIGURE 8. Comparison of key accuracy metrics when one robot is deployed. The results on the left involve the UEs that benefit localization ($t_{lead} > 0$) with geometrical constraints of *angles_range* = [35, 170]. The distribution is then compared for different ranges, with the boxplots on the right.

victims positioned at the beginning of the operation in the lower-left corner—the robot is unable to collect sufficient information to execute the algorithm and, instead, reaches the victim directly. Similarly, for victims located on the left side of the map, the AoA intersection angle of received signals remains below 35 degrees, as the robot follows a vertical trajectory from the lower-left to the upper-left corner. The last group of victims that do not benefit from RF-based localization are those situated in the upper-right quarter of the map, completely obstruct by the obstacles.

The fourth plot depicts the *Leading Localization Time* (t_{lead}), representing how early a victim is localized before being accessed in the baseline exploration. This value is simply the difference in seconds between the baseline target access time and the localization time. Excluding 21 victims placed in the state x_0 that are automatically accessed at the beginning of the operation, results indicate that 182 cases (38%) are detected in advance. These are exclusively located in the right half of the map, according with the considerations made for the localization time. Among these, 18% have t_{lead} below 30 seconds, 44% between 30 and 60 seconds, 12% between 60 and 90 seconds, and 26% exceeding 90 seconds.

The *Leading Localization Time* allows for dynamic path adjustments, enabling earlier victim access than initially

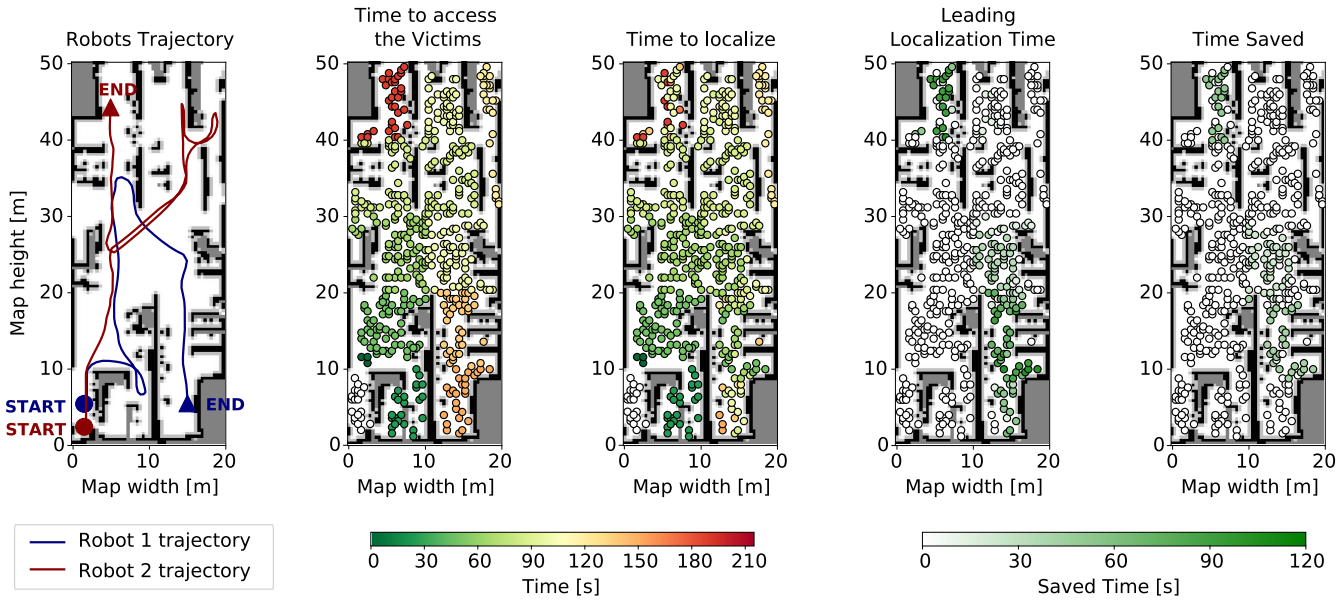


FIGURE 9. Comparison of key time metrics when two robots are deployed.

Error distributions along different intersection angle ranges (2 robots):

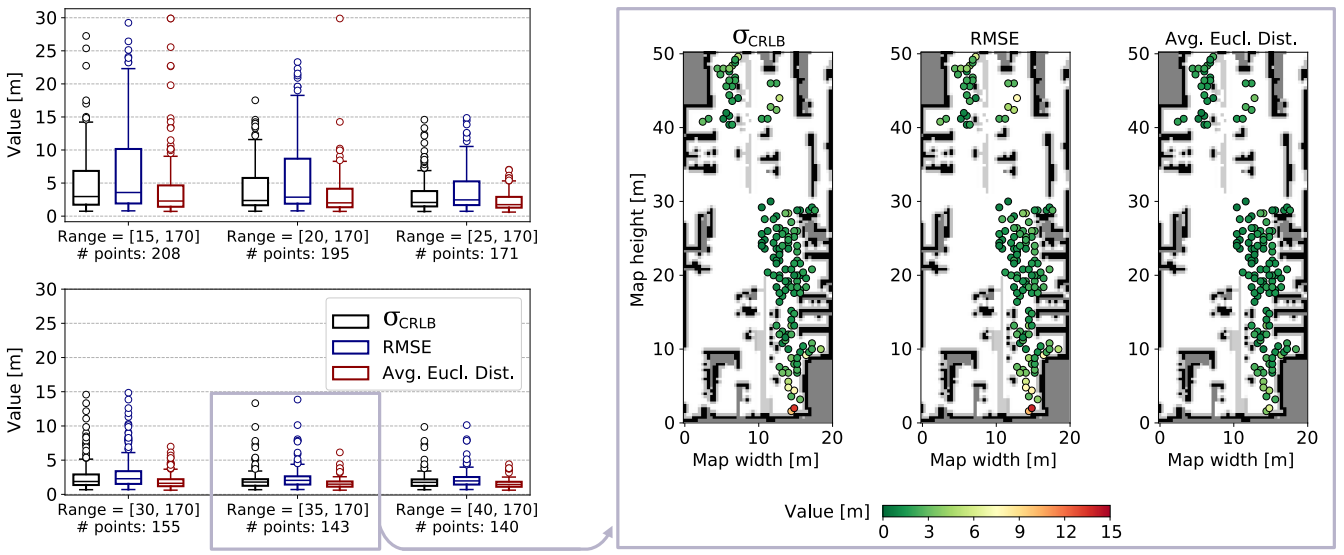


FIGURE 10. Comparison of key accuracy metrics for the UEs that benefit localization when deploying two robots.

planned and leading to an overall reduction in rescue time. This improvement is quantified by the *Time Saved* t_{saved}, which registers benefits for 29% of the victim population. Specifically, 35% of these cases show a time reduction of 25-30 seconds, 52% experience a reduction of 40-60 seconds, and 13% benefit from reductions of over 80 seconds.

Figure 8 visually presents the values of each localization performance metric on the 2D map, showing only the targets for which remote position estimation led to positive t_{lead}. The results confirm that the selected *angles_range* $= [35°, 170°]$ provides acceptable localization performance, with accuracy below 5 meters for most of the points, for each metric. Specifically, over the 182 occurrences, the 87.2% register a squared root trace below 5 meters while the remaining 12.8% are between 5 and 10 meters. Analogously for the RMSE, with slightly worse values and 3 occurrences exceeding 10m. The distribution of these values is shown in the box plots on the right side of the figure. Here are also compared the same performances with different *angles_range*, and it is possible to notice that suboptimal geometrical settings lead to suboptimal accuracies, although affecting more points and with better timing. A more stringent range (e.g., [40,170]) provides better but minimal improvements at the price of fewer targets and more time to triangulate them. Analogously to earlier figures, metric values exceeding the maximum scale (30 meters for the boxplots) have been clipped to improve

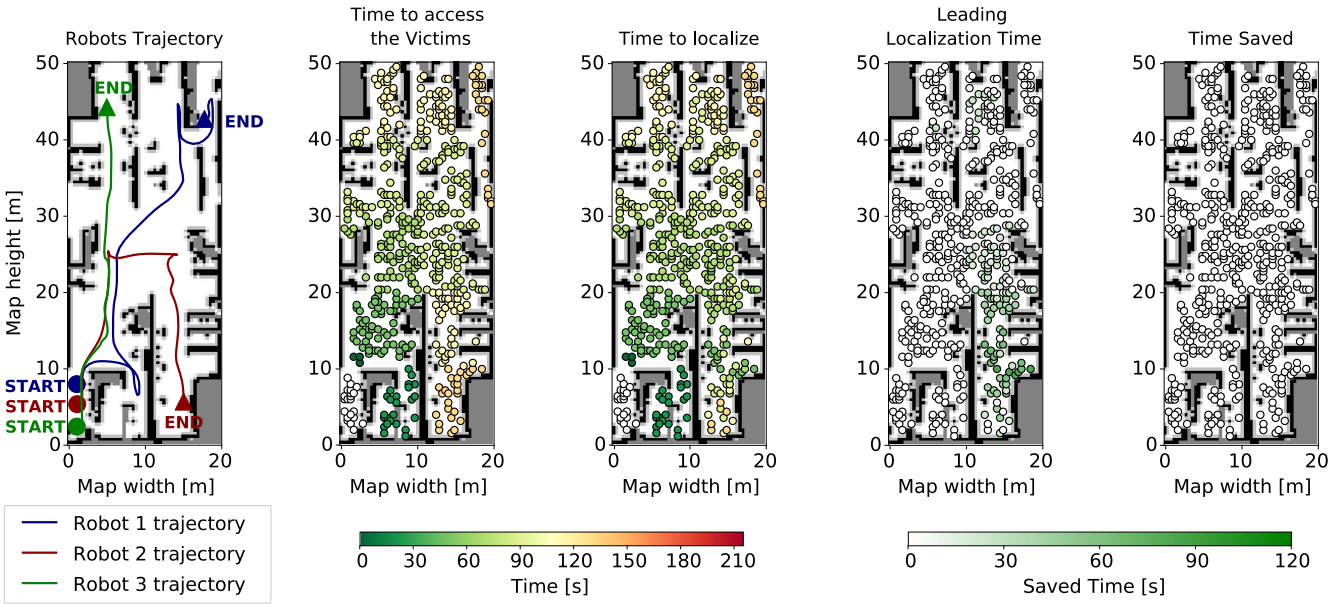


FIGURE 11. Comparison of key time metrics when three robots are deployed.

Error distributions along different intersection angle ranges (3 robots):

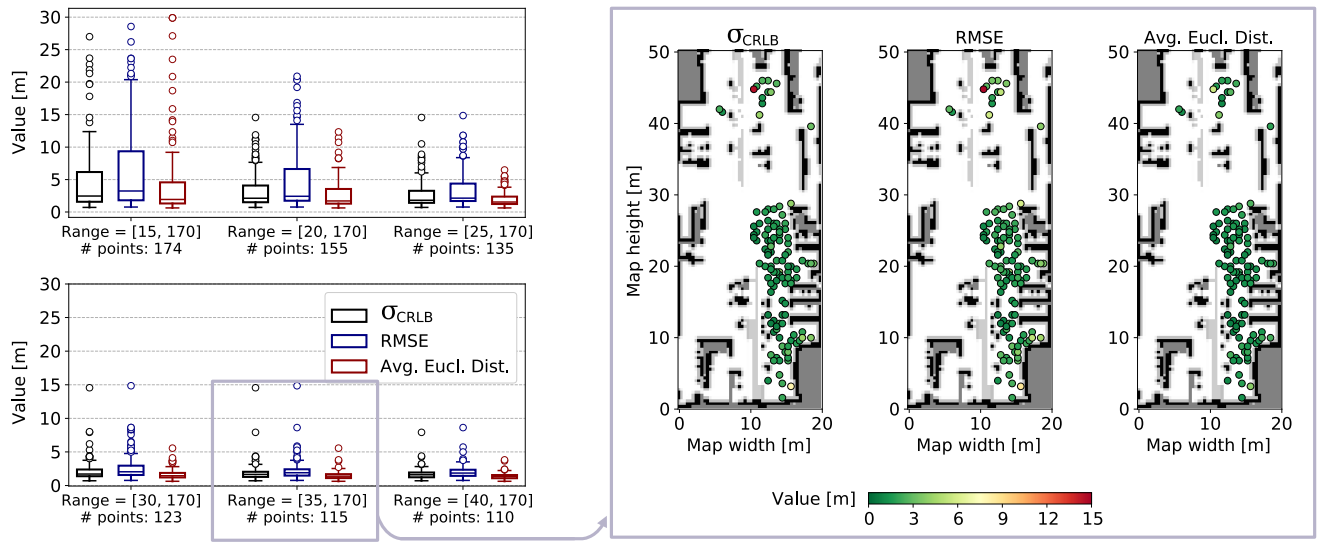


FIGURE 12. Comparison of key accuracy metrics for the UEs that benefit localization, when three robots are deployed.

visual clarity and focus on the more informative range of the distribution.

Similar considerations are made for the deployments with two and three robots, as shown in Figures 9 to 11, respectively. In general, the localization times tend to improve as the number of deployed robots increases. This is attributed to the faster spatial coverage of the environment, which allows robots to satisfy the geometric constraints for accurate localization earlier. However, a better t_{loc} does not necessarily translate into consistent gains across all metrics when compared to the single-robot scenario, as multiple robots accelerate exploration and reduce the average time to encounter victims in the baseline approach. Reaching all areas faster on the map diminishes the relative advantage of early victim detection in path planning. Specifically, in the case of two robots, 29.8% of the victims (143 occurrences) benefit from a positive *Leading Localization Time* (t_{lead}), while executing the trajectories proposed in Figure 9. Specifically, among these cases, 19% fall within 0 and 30 s, 44% between 30–60 seconds, 16% between 60 and 90 s, and 21% exceed 90 s.

Regarding *Time Saved* (t_{save}), 17.7% of the victims benefit from a reduction in access time. Among them, 49% experience a time reduction of 22 s, 30% by 42 s, and 21%

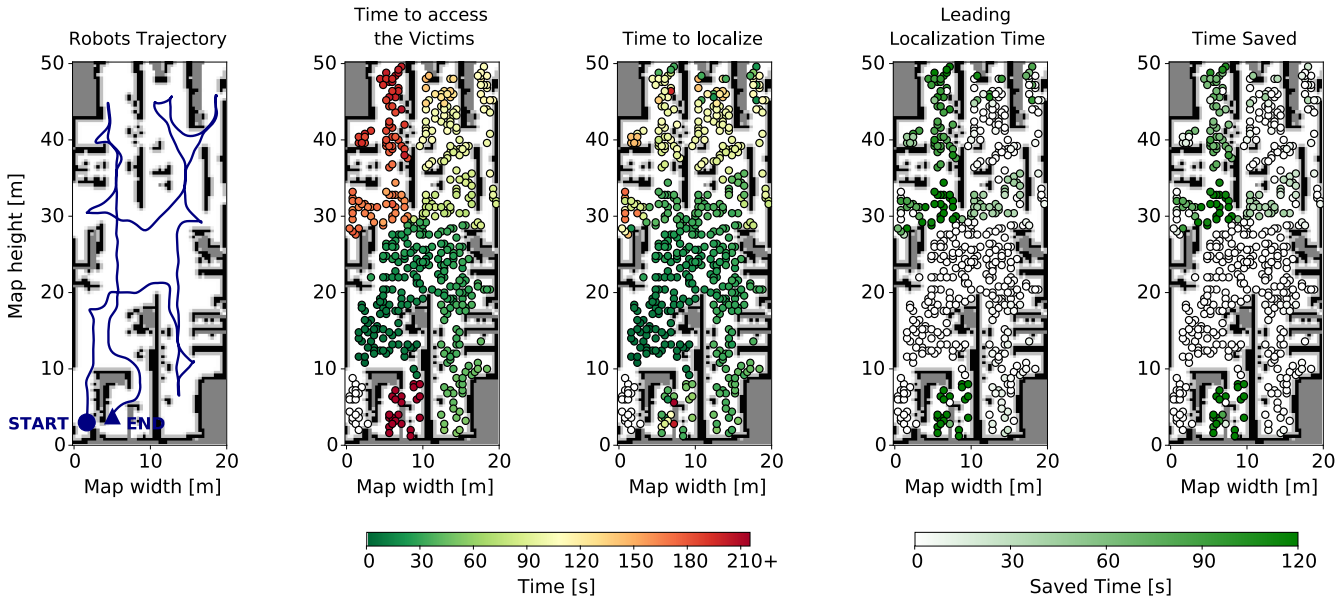


FIGURE 13. Comparison of key time metrics for a single-robot deployment with the online TSP-based exploration.

by 51 s, demonstrating a clear advantage of the proposed approach.

For the three-robot deployment, 24% of the victims (115 occurrences) experience a positive t_{lead}, with 59% within 30 s, 27% are between 30 and 60 s, 9.5% are between 60 and 90 s, while only 4.5% are exceeding 90 s. However, in this scenario, t_{saved} is effectively zero, as the three robots distribute optimally across the map. Their trajectories ensure that each area is covered efficiently and in the fastest way already. As for the localization accuracy, the distribution of the metrics scores is reported with the plots in Figures (Fig. 10 and 12). For over 95% of occurrences, the localization precision is within 5 meters, only one instance in each configuration exceeds a 10+ meters error; in the remaining cases are marked with suboptimal accuracy. The accuracy studied in relation to different *angles_ranges* is similar to that analyzed in the single robot scenario for both two and three-robot deployments.

D. OPERATION TIME SAVINGS FOR TSP-BASED EXPLORATION

To assess the impact of localization on TSP-based exploration, we conducted experiments under two conditions: first using the standard TSP-based approach, and then employing the localization-enhanced variant. Figure 13 illustrates the trajectory of a single robot, truncated when full coverage of the map is achieved (the return to the start is omitted, as exploration is complete).

The second subplot reports the time at which each target position is covered within the robot's LiDAR sensing radius (8 m). The remaining plots present the key temporal metrics, namely localization time and saved time relative to the baseline schedule. The latter highlights the following improvements: the target was accessed earlier than scheduled in the baseline TSP execution in 162 cases, corresponding to $\sim$ 34.0% of the total (excluding the 20 targets automatically accessed at time zero due to the 8m sensing radius). The distribution of time savings is as follows: 26.0% of improvements are within 30 seconds, 22.2% between 30–60 seconds, 25.3% between 60–90 seconds, 9.2% between 90-120 seconds, and 17.3% exceed 120 seconds. These results confirm that localization can accelerate access to a substantial fraction of targets, with non-negligible gains in several cases.

Figure 14 shows results with two robots. Similarly, the trajectory is truncated upon full coverage. In this case, localization-aware replanning enabled 118 earlier accesses out of 477 targets (24.7%), among which 73.7% of time gains were within 30 seconds, 24.6% within 30–60 seconds, and 1.7% between 60–90 seconds. Similarly to the first benchmark, the improvements with three-robot deployment (Fig. 15) are limited by the high coverage efficiency achievable with multiple agents in the map under test. The 18.0% of targets accessed earlier have an average time saving of $\sim$ 11 seconds.

VIII. FUTURE WORKS

While this study focuses on single-victim localization per experiment, an important next step is to extend the framework to scenarios involving multiple victims that must be simultaneously identified and reached. Handling multiple victims introduces significant challenges, particularly managing interference and ambiguity in signal processing. The current localization method based on received signal strength (RSS) and AoA estimation may require enhancements such as multi-source signal discrimination, robust filtering techniques, or probabilistic data association strategies.

Another promising direction is integrating predictive models into local motion planning, where the model predicts

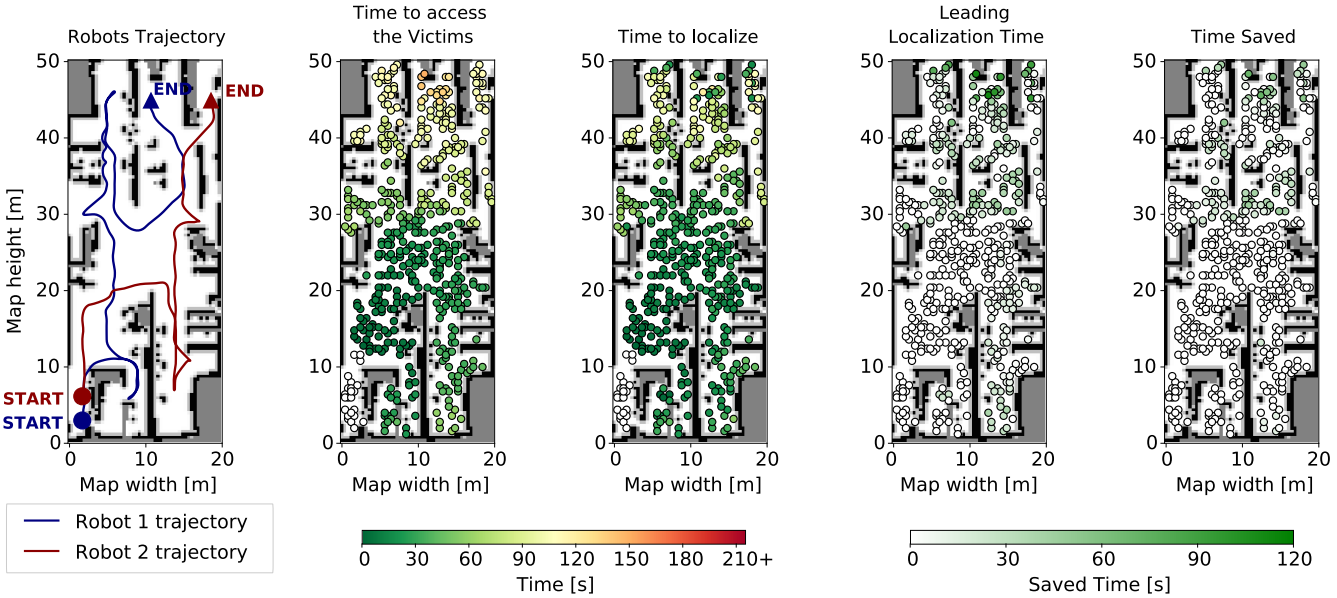


FIGURE 14. Comparison of key time metrics when two robots are deployed with the online TSP-based exploration.

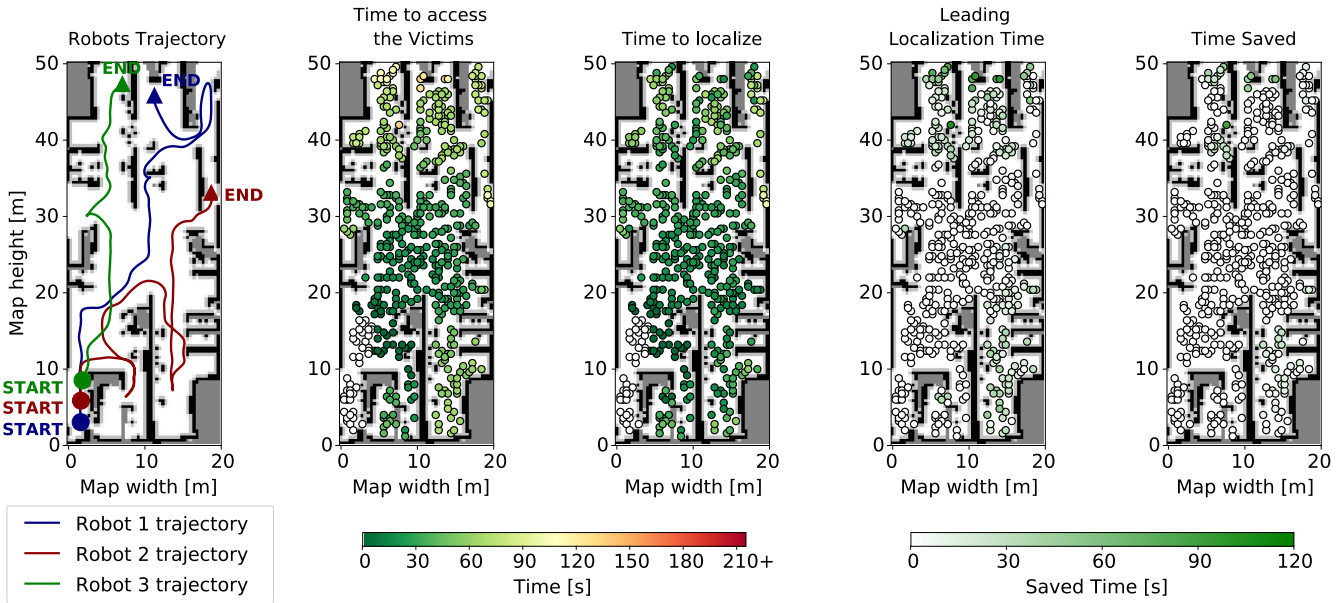


FIGURE 15. Comparison of key time metrics when three robots are deployed with the online TSP-based exploration.

the robot's next optimal state to reduce uncertainty in the target's estimated position.

Lastly, incorporating vertical transitions and floor-level semantics would significantly increase the planner's applicability in complex real-world scenarios such as multi-story office buildings or industrial facilities. To support multi-floor indoor environments, the global planner should maintain multiple grid-based representations (one per floor) and model the connectivity between floors. While the global planner would require explicit extensions to handle multi-floor environments, the local planner architecture is already agnostic to floor levels; scaling to multi-floor scenarios at the local level is as straightforward as scaling to larger single-floor environments, requiring no structural modifications to the local planning module.

IX. CONCLUSION

Indoor Search and Rescue (SAR) operations using ground robots have often overlooked integrating real-time localization into exploration strategies, in contrast to Unmanned Aerial Vehicle (UAV)s operating in outdoor environments that present different challenges. This paper presents a novel integration of remote victim localization into multi-robot exploration for indoor SAR. The probabilistic localization, based on Angle of Arrival (AoA) triangulation and refined through Received Signal Strength Indicator (RSSI) filtering,

is validated using state-of-the-art metrics such as Cramér-Rao Lower Bound (CRLB) and Root Mean Square Error (RMSE). This information is used to prioritize exploration toward areas with expected victims. Extensive simulations demonstrate improvements of up to 28% faster victim reach and 50% reduction in operation time in indoor scenarios, compared to existing methods.

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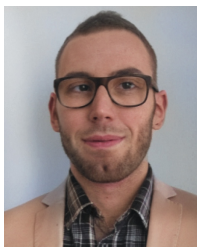
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